Pergamon

Available online at www.sciencedirect.com

SCIENCE DIRECT®

Acta Materialia 51 (2003) 741–752

Acta MATERIALIA

www.actamat-journals.com

Temperature induced porosity in hot isostatically pressed gamma titanium aluminide alloy powders

Gerhard Wegmann, Rainer Gerling *, Frank-Peter Schimansky

GKSS Research Centre, Institute for Materials Research, Max-Planck-Strasse, 21502, Geesthacht, Germany

Received 13 July 2001; received in revised form 26 August 2002; accepted 25 September 2002

Abstract

In this study the occurrence of temperature induced porosity (TIP) in hot isostatically pressed (HIP) compacts of different gamma Titanium aluminide alloys was investigated. Two gamma Titanium aluminide alloys Ti-48.9at.%Al and an advanced Niobium containing alloy Ti-46at.%Al-9at.%Nb have been atomized by gas atomization and by centrifugal atomization in an inert gas atmosphere. The alloy powders were studied regarding porosity and the content of inert gas entrapped in the powder particles. Selected powder batches were hot isostatically pressed at 1280 °C and were investigated with respect to TIP evolution after a high temperature exposure to 1390 °C for short and long time periods. It was found that gas atomized Titanium aluminide alloy powders contain a certain amount of atomization gas, the concentration of which increases with the powder particle size. The amount of inert gas entrapped in centrifugally atomized powders is higher as compared to powders produced by gas atomization. The occurrence of TIP after high temperature annealing of the HIP' ed compacts depends on the grain size, the processing medium (Argon or Helium), the amount of entrapped inert gas and the annealing time. Guidelines are presented for minimizing or prevention of TIP in γ-TiAl alloys processed by powder metallurgy.

Keywords: Hot isostatic pressing (HIP); Intermetallic; High temperature annealing; Temperature induced porosity (TIP)

1. Introduction

After an intensive research period of about 15 years, several γ-TiAl alloys have been successfully developed [1–4]. For successful commercialization more emphasis is to be put on the processing of the existing alloys in order to promote the production of structural components economically and competitively to other materials. The use of γ-TiAl alloys is mainly restricted to elevated temperature applications in air propulsion systems, e.g. blades, vanes and disks in the low and high pressure compressor stages of gas turbines. γ-TiAl alloys provide a high specific strength, a high specific Young's modulus and creep and oxidation resistance up to temperatures of 700 °C. The strength retention with temperature outperforms that of conventional Titanium alloys. The high specific strength of γ-TiAl alloys accounts for their potential to partly replace heavy Nickel superalloys in

* Corresponding author. Fax:+49-4152-87-2534.
E-mail address: rainer.gerling@gkss.de (R. Gerling).

1359-6454/03/$30.00
doi:10.1016/S1359-6454(02)00465-2

the hot sections of the gas turbine compressor. Against this background a gas atomization facility, especially designed for the production of clean Titanium and Titanium Aluminide powders, was installed at GKSS Research Centre some years ago [5,6]. The processing of γ-TiAl alloys by powder metallurgy can offer several advantages with respect to the ingot route [7,8]. Because of the double peritectic reaction in case of the advanced γ-TiAl alloys with Al-additions of 44–49at.% a slow solidification, like that in a cast ingot, always results in severe macrosegregations and corresponding inhomogeneous microstructures [9]. Besides that, cast γ-TiAl alloys show a coarse microstructure, which has to be refined by subsequent processing steps. With P/M processing of γ-TiAl alloys very fine and chemically homogeneous microstructures can be achieved in a single compaction step like hot isostatic pressing (HIP) [3,8,10,11]. Several projects have been conducted using PM processed γ-TiAl alloys, e.g. a feasibility study for the manufacturing of hollow turbine blades using sheet material [12]. An advanced sheet-rolling route has been developed by Plansee AG, Reutte, Austria, to roll sheets with a thickness down to 1 mm [3,4,13–15]. The prematerial for the sheet was initially HIP-consolidated alloy powder. The powders were produced by Argon gas atomization. The sheets were found to contain pores [3,4,14], which were aligned in rolling direction and could give rise to some degradation of the mechanical properties, e.g. creep resistance. No adverse effect of the porosity on high cycle fatigue could be detected [3,14]. Argon gas, which was entrapped during atomization and probably stored in micro-pores, is suspected to be the origin of the observed porosity. During a high temperature exposure, e.g. during rolling, the Argon pores grow substantially by the expansion of the enclosed Argon resulting in “so-called” temperature induced porosity (TIP) [3,14].

For gas atomization of γ-TiAl alloys only inert gases such as Argon or Helium can be used because of the highly reactive melt. However, during atomization the liquid metal interacts with the inert gas environment, leading to the entrapment of gas within the particles in spherical gas pores [16–21]. In a recent study on the Argon atomized alloy Ti-47Al-4(Mn, Nb, Cr, Si) (at.%) it was shown that the internal porosity of alloy powders increases with the powder particle size and ranges up to 1.0vol.% [17]. Inputting the measured porosity in a simple model the Argon content of different powder size fractions was calculated. The good correlation of the calculated and analyzed Argon concentrations indicated that nearly all the observed pores were filled with Argon gas [17]. In earlier work Eylon et al. [22] showed clearly that HIP-consolidated REP-powders of Ti-6Al-4V (REP: rotating electrode process) occasionally suffer from TIP of up to 1.0vol.% under certain conditions. However, Eylon et al. were able to demonstrate that TIP did not cause any reduction in fatigue life [23]. In the case of the conventional Titanium alloys, P/M processing, i.e. HIP of alloy powders, is regarded as an established technique for the production of high quality complex shapes for aerospace applications. The mechanical properties of these components are known to be equal or better than cast and wrought material [24].

In recent work [16] we showed that binary Ti-48.9at.%Al powder produced by two different gas atomization techniques using Argon and by centrifugal atomization in an Argon atmosphere exhibited a certain portion of Argon porosity, the level of which varied with the atomization technique. On the other hand the Argon porosity almost entirely vanished by the HIP treatment and no reopening of the Argon pores resulted as a consequence of a high temperature exposure for either of the atomization techniques. The subject of the present study was to elucidate different influences on the phenomenon of gas entrapment and TIP predominately for P/M-processed gamma Titanium Aluminide alloys. In detail, four variables have been investigated, which were: i) the powder particle size, ii) the atomization technique, iii) the atomization medium: Argon or Helium, and iv) the alloy system in order to elucidate, whether there are differences between the behavior of the well-known conventional Titanium alloys and Titanium Aluminide alloys on the one hand and whether there are also differences between different γ-TiAl alloys on the other hand. The experiments were carried out with two γ-TiAl alloys, a binary model alloy Ti-48.9at.%Al (TiAl35; this nomenclature corresponds to the Al content in wt%) and an advanced Nb containing alloy Ti-46at.%Al-9at.%Nb (TNB) and, for comparison, on a conventional Titanium alloy Ti-6wt.%Al-7wt.%Nb (Ti-67), which is basically the biocompatible version of the widely used Ti-6wt.%Al-4wt.%V alloy.

2. Experimental details

The alloys for this study were all produced in an identical way starting with the elemental constituents molten by means of a plasma torch in a water-cooled copper crucible. The only exception is the alloy Ti-67, atomized with the EIGA-technique (Electrode Induction Melting Gas Atomization), for which commercially purchased rods were used as the starting material. The electrode rods of the γ-TiAl alloys for the EIGA-technique were prepared by skull melting with a plasma torch and casting into a water-cooled copper mould reflecting the shape of the EIGA-rods.

The powders were produced by three different atomization techniques, designated as PIGA (Plasma Melting Induction Guiding Gas Atomization), EIGA and CA (Centrifugal Atomization). The PIGA and EIGA powders were produced with Argon as the atomization gas only, while the CA experiments for the two γ-TiAl alloys were performed with either Argon or Helium as the cooling gas (Table 1).

In the case of PIGA (Fig. 1a) the alloy is skull melted by a plasma torch. The melt runs—guided by a specially designed induction heated guiding system—into an annular ring slit nozzle of ‘Laval’-type, where the melt is disintegrated into fine droplets. The nozzle is operated at a pressure of 2.5 MPa of Argon. Concerning the application of the EIGA-technique (Fig. 1b), an alloy electrode is melted by an induction coil and a more or less continuous melt stream enters a conventional ring slit nozzle (parallel slit). The EIGA-nozzle is operated at the same pressure as the PIGA-nozzle; however, the opening diameter is only about 2/3 that of the PIGA-nozzle. This together with the different nozzle design could possibly result in differences of the powder characteristics, specifically its porosity and Argon concentration. In the case of CA (Fig. 1c) the prealloyed button is melted by the plasma torch and the melt is poured onto a rotating copper disk. The disk rotates initially at a speed of 18.000 rpm in an inert gas atmosphere. Compared to PIGA and EIGA the CA process does not require such large quantities of expensive processing gas.

Table 1
Matrix visualizing the powders produced within this study, atomization techniques, gases and alloys (*x*: conducted, –: not conducted)

Alloy	PIGA Ar	EIGA Ar	CA Ar	CA He
TiAl35	x	x	x	x
TNB	x	x	x	x
Ti-67	x	x	–	–

For the investigation of the porosity and the Argon concentrations in the powders, four powder size fractions (32–45, 63–90, 125–180, 250–355 μm) in the case of the PIGA-powders were extracted from each powder batch by sieving. For the EIGA-powders the coarsest size fraction 250–355 μm was practically non existent, while for the CA-powders the finest size fraction was lacking due to the characteristic particle size yields described in section 3.1 (Fig. 2). In addition, alloy powders of the two γ-TiAl alloys, TiAl35 and TNB, of the size fraction 125–180 μm (powders of all production techniques) were filled into Titanium cans and degassed under vacuum ($p \leq 10^{-3}$ Pa) at 400 °C. Subsequently, the cans were sealed gas tight and HIP-compacted at 1280 °C for 4 h under a pressure of 200 MPa. The HIP-compacts were investigated with respect to porosity and Argon concentration. Samples of 10 mm height and Ø 16 mm diameter of the HIP-compacts were heat treated at 1390 °C for either 0.3 h or for 4 h in order to study TIP evolution of the various compacts. These heat treatments generated a creep resistant fully lamellar microstructure.

Powders and compact samples were embedded in a graphite resin for optical investigation of the porosity. The specimens were mechanically ground and polished using standard procedures. The porosity was determined by means of plane section

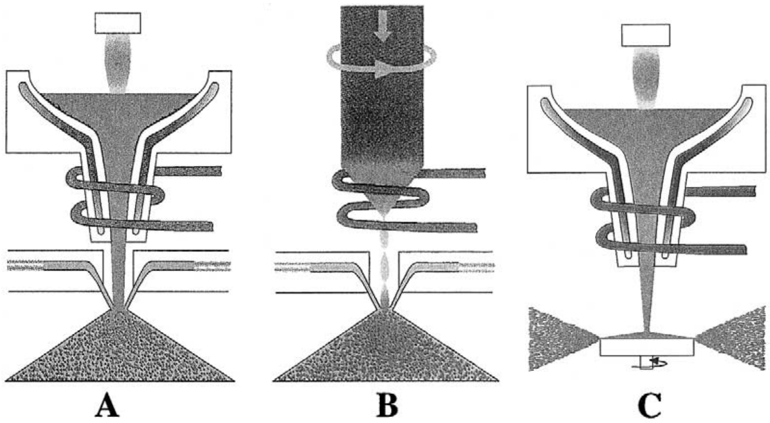

Fig. 1. Schematic illustration of the atomization techniques. A: Plasma Melting Induction Guiding Gas Atomization (PIGA), B: Electrode Induction Melting Gas Atomization (EIGA), C: Centrifugal Atomization (CA).

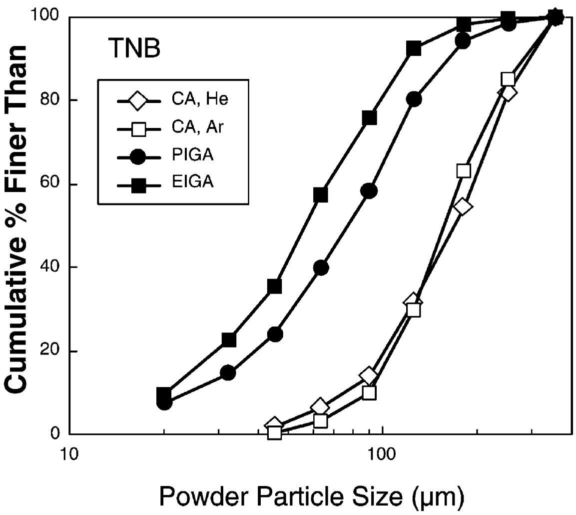

Fig. 2. Powder particle size distributions for various atomization techniques for the TNB alloy.

analyses (PSA) using a combination of a Zeiss Axioplan optical microscope and an image analysis system. From each powder size fraction at least 2000 powder particles were investigated. The magnification was adjusted suitably to the particle size. Only pores that were seen to be closed in the plane of section were counted for the porosity evaluation. The smallest pore size accessible by the optical microscope was given by its resolution: 1μm. The porosity of the compacts was determined at a constant magnification of 50× accounting for the smallest detectable pore to be 2 μm in diameter. An area of at least 45 mm^2 was evaluated for a single condition and the data were treated according to standard statistical procedures, i.e. the standard deviation of 48 discrete measurements was taken as the relevant error.

The Argon concentration measurements were performed by Böhler Edelstahl GmbH, Kapfenberg, Austria, using a combination of a carrier gas based fusion instrument as an extraction system and a quadrupole mass spectrometer as the detection system. The accuracy of this method in terms of the standard deviation was 0.10 μg/g for compact samples and 0.15 μg/g for powder samples. The Helium concentration of the Helium processed powders could not be analysed.

3. Results

3.1. Powder particle size distribution

Each atomization technique results in a characteristic particle size distribution, as for example shown for TNB in Fig. 2. Comparable curves for the other investigated alloys were obtained. The EIGA-technique produces the finest powder with a d_{50} = 55 μm. The PIGA-technique gave a slightly coarser size distribution with respect to the EIGA-technique with a d_{50} = 77 μm. The CA technique, employed either in an Argon or a

Helium atmosphere, results in a much coarser powder with a $d_{50} = 150$ µm (Argon) and $d_{50} = 160$ µm (Helium). For this consideration only the powder sieved to below 355 µm was analysed. It must be pointed out that the CA-technique has not been optimized with respect to a high yield of fine powder. The optical micrographs in Fig. 3 display that the PIGA and the EIGA-techniques (Fig. 3a and b) result in almost perfectly spherical particles with almost no agglomeration. However, the powder produced by the CA-technique (Fig. 3c and d) shows a certain fraction of irregularly shaped particles, which increases with the particle size, i.e. the small size fractions < 90 µm of the CA-powders are almost entirely spherical in shape like the PIGA and EIGA-powders.

3.2. Porosity in powder particles

It has been shown in previous studies using PIGA- atomized TiAl35 [16] and other advanced γ-TiAl alloy powders [17,18], that the powder porosity is higher, the larger the powder particles are. For TiAl35 such a correlation could also be found for the EIGA- and CA-atomization techniques [16]. The new results obtained for intermetallic TNB and the titanium alloy Ti-67 show the same tendencies (Table 2).

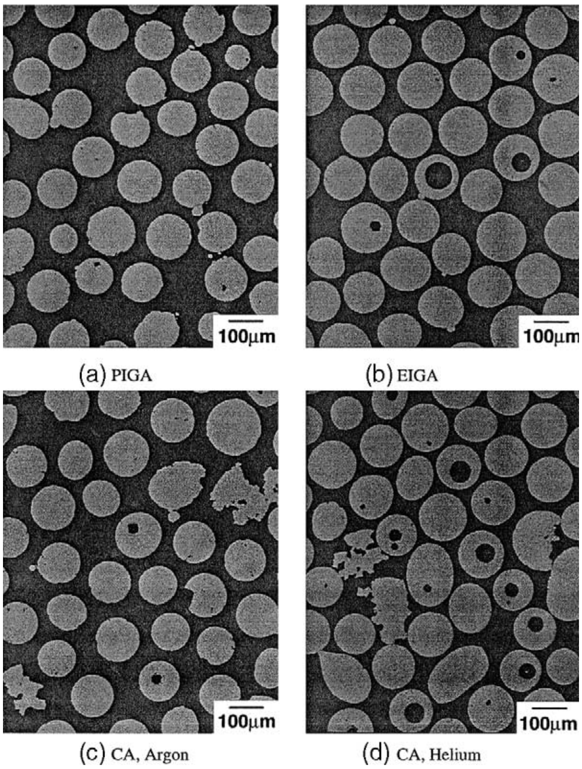

Fig. 3. Optical micrographs of the TNB-powder size fraction 125–180 µm.

In addition, systematic differences in the powder porosities for different atomization techniques become visible. PIGA- and EIGA-powders offer considerably less porosity than the CA-powders over all size fractions investigated. In general PIGA-powders reveal less porosity compared with EIGA-powders (Table 2). This is clearly seen for TNB and Ti-67, however, the alloy TiAl35 does not show this distinct tendency for the fine fractions 32–45 µm and 63–90 µm. CA-powder production under an Argon atmosphere leads to a smaller porosity compared with the CA-production under a Helium atmosphere.

For identical powder particle size fractions and for powders fabricated with the same atomization technique the porosities are comparable for the three alloys investigated (Table 2). This means that the gamma Titanium Aluminide alloys show comparable porosity levels with respect to the conventional Titanium alloy Ti-67 and among themselves, i.e. there is no alloy effect discernible within the range of this study.

3.3. Argon content

In agreement with the increase of porosity with the powder particle size, the Argon content increases with the particle size for the PIGA- and EIGA-powders, while no such tendency is observed for the CA-powders (Fig. 4). In the case of the CA-powders, surprisingly, the largest powder size fraction contains the least amount of Argon, even though the porosity is the highest in the largest powder particles.

As already seen with porosity, the PIGA- and EIGA-powders contain less Argon as compared with the CA-powders. In agreement with the porosity results the Argon content of the PIGA-powders is for all three alloys investigated lower than that of the EIGA-powders.

For PIGA- and CA-powders there may be an

Table 2
Porosity data measured according to the plane section method in an optical microscope on at least 2000 powder particles (given in vol.%)

Alloy	Atomization	32–45 μm	63–90 μm	125–180 μm	250–355 μm
TiAl35	PIGA	0.12	0.16	0.22	0.28
	EIGA	0.06	0.10	0.47	–
	CA Ar	–	0.39	0.68	0.94
	CA He	–	0.39	2.14	2.64
TNB	PIGA	0.03	0.12	0.26	0.52
	EIGA	0.08	0.14	0.52	–
	CA Ar	–	0.28	0.39	1.73
	CA He	–	0.26	0.63	2.06
Ti-67	PIGA	–	0.06	0.25	0.21
	EIGA	0.06	0.16	0.53	–

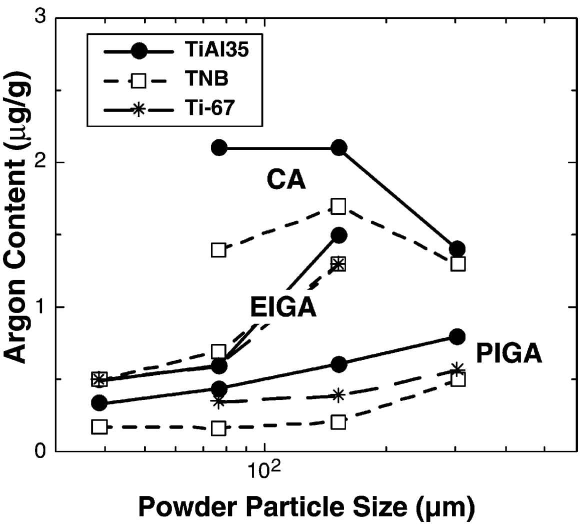

Fig. 4. Argon content versus powder particle size as a function of the various atomization techniques.

alloy effect on the tendency for Argon entrapment from the data (Fig. 4), namely that the advanced alloy TNB captures less Argon compared with the binary model alloy TiAl35. In the case of the EIGA-powders no alloy effect is discernible from the data obtained, all three alloys capture more or less the same amount of Argon gas during atomization.

As expected, the results displayed in Table 3 indicate that the Argon concentration in the HIP-compacts is more or less the same as that of the corresponding powders within the range of error.

3.4. Effect of high temperature exposure on the porosity of HIP-compacts

After HIP of the powder size fraction 125–180 μm, a fine near γ microstructure was obtained in the various compacts of TiAl35 and TNB (as an example Fig. 5a). The HIP-compacts of the PIGA-powders exhibit a γ grain size of 15 μm (TNB) and 25 μm (TiAl35), and in the case of EIGA-pro-

Table 3
Argon concentration before and after HIP-consolidation of the powder size fraction 125–180 μm (given in μg/g)

Alloy	Atomization	Powder (125–180 μm)	HIP-compact
TiAl35	PIGA	0.6	0.5
	EIGA	1.5	0.9
	CA, Ar	2.1	2.3
TNB	PIGA	0.2	0.4
	EIGA	1.3	1.0
	CA, Ar	1.7	1.7

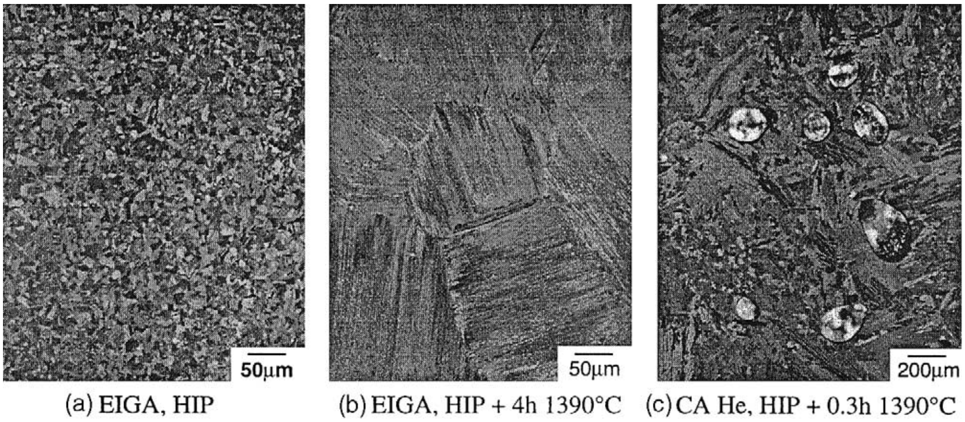

Fig. 5. Microstructures of TNB-compacts prepared with EIGA-powder of the size fraction 125–180 μm a) before and b) after thermal exposure, and c) with CA-Helium powder (125–180 μm) after thermal exposure, average porosity=2.6vol%.

cessed powders a comparable grain size of about 15 μm resulted for TNB (Fig. 5a) and TiAl35. After a high temperature exposure of 0.3 h at 1390 °C a fully lamellar microstructure developed in all compacts with colony sizes of 200 μm (TNB) (Fig. 5b) and around 1 mm (TiAl35). A prolonged heat treatment of 4 h at 1390 °C resulted in some coarsening of the colony size in the case of TiAl35, but no significant coarsening of the colony size in the case of TNB.

3.4.1. TiAl35 alloy powder

The effect of high temperature exposure on the porosity of the HIP-compacts is illustrated in Fig. 6a (TiAl35) and Fig. 6b (TNB). For the binary alloy TiAl35, no TIP after annealing at 1390 °C was detected in compacts made from PIGA and EIGA powders; the porosity levels remained below the detection limit of 0.01vol.%. For all as HIP′ ed compacts investigated within the framework of this study the measured porosity values ranged below 0.01vol.% and this value was defined as the detection limit, i.e. a measured porosity value of 0.01vol.% or less is regarded as no porosity throughout this article. The compact made from centrifugally atomized TiAl35-powder, processed with **Argon** (CA, Ar) and heat treated at 1390 °C for 4 h, shows numerous pores 5–10 μm in diameter within the plane of section. The largest observed pore was actually 64 μm in diameter. However, pores in the as HIP′ ed condition of TiAl35 were rarely observed and always below 35 μm. The average porosity of the CA,Ar-compact after a high temperature exposure for 4 h was determined to be 0.04vol.%, however, no porosity was encountered after a short time exposure to 1390 °C for 0.3 h. The compact fabricated from centrifugally atomized TiAl35-powder, processed with **Helium** and heat treated at 1390 °C exhibits significant TIP. After a heat treatment time of 0.3 h an average porosity of 0.22vol.% was determined and with an extension of the exposure time to 4 h the average porosity increased up to 0.45vol.%. Pore sizes of up to 120 μm (0.3 h, 1390 °C) and 260 μm (4 h, 1390 °C) were observed.

3.4.2. TNB alloy powder

In the case of the Niobium containing alloy TNB, TIP was measured in the compacts of all powder processing techniques after sufficient exposure times more or less pronounced. After HIP, the compacts of all TNB-powders did not show porosity. After a temperature exposure to 1390 °C for 0.3 h the compact made of the **Argon**-atomized PIGA powder did not show significant porosity, but after an extension of the exposure time to 4 h the average porosity was determined to be 0.05vol.%. For the compacts made from EIGA- and CA,Ar-powders some porosity was already detected after a short high temperature exposure of 0.3 h, the average measured porosity levels were 0.02vol.% and the largest observed

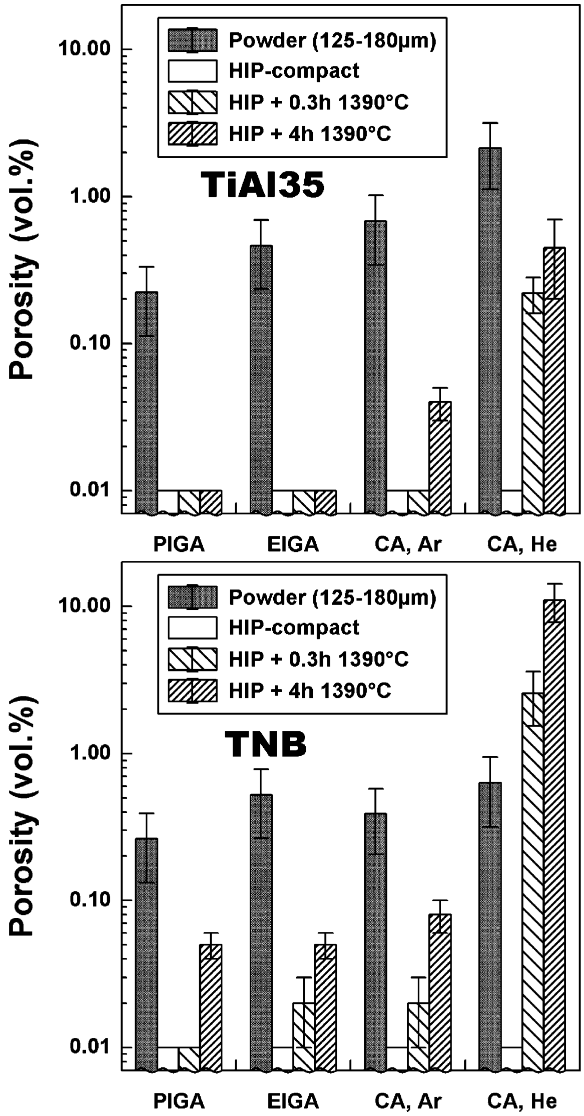

Fig. 6. Temperature induced porosity (TIP) evolution of TiAl alloy powders (powder size fraction 125–180 µm) consolidated by hot isostatic pressing (HIP) after high temperature exposure to 1390 °C. The powders of two alloys TiAl35 and TNB were produced each by different techniques as indicated in the figure.

pore sizes were 31 µm (EIGA) and 61 µm (CA,Ar). After the extended high temperature exposure (4 h, 1390 °C) even more pores were observed and the average porosity increased to 0.05vol.% (EIGA) and 0.08vol.% (CA,Ar). Like the alloy TiAl35, the TNB alloy centrifugally processed with **Helium** shows the most serious susceptibility for TIP. After a high temperature exposure of 0.3 h at 1390 °C the porosity increased to 2.6vol.%. The typical pore appearance after an exposure to 1390 °C for 0.3 h in the compact made from CA-powder processed with Helium is shown in Fig. 5c. The Helium-pores have a spherical shape and their size ranges up to 300 within the plane of section. After an extended exposure to 1390 °C for 4 h a high number of very large pores had evolved sometimes several millimeters in size with interconnecting channels. The average porosity was determined to 11.0vol.%.

It should be noted that in all compacts (TiAl35 and TNB) showing TIP, the porosity was exposure time dependent.

4. Discussion

This study confirmed that temperature induced porosity (TIP) is clearly an issue in P/M-processed γ Titanium aluminide alloys like it is for conventional Titanium alloys, as shown by Eylon et al. [22,23]. No significant difference between conventional Titanium (Ti-67) and γ Titanium aluminide alloy powders was found regarding porosity and Argon gas entrapment (Table 2 and Fig. 4). However, the present results show that the undesired phenomenon of TIP in HIP-consolidated gamma Titanium aluminides can be minimized or even prevented by alloy selection and by a proper choice of the atomizing technique, the atomizing medium and the annealing time. In the following, two factors will be discussed separately, these are firstly the microstructure in the HIP-compacts at the annealing temperature in terms of different grain sizes and secondly the amount and type of inert gas stored in the powder particles.

4.1. Effect of the microstructure and the annealing time

The significant difference of the TIP-susceptibility between the binary alloy TiAl35 and the Niobium containing alloy TNB (Fig. 6) is suspected to be a consequence of the different microstructures. While the binary alloy TiAl35 is in the α single-phase field at the annealing temperature of 1390 °C, the TNB alloy is (at the same temperature) already in the $\alpha + \beta$ two-phase field. If we presume that the resulting colony size after cooling tells us approximately the grain sizes at the annealing temperature (the cooling rates were identical) we can conclude that the TNB alloy provides a five times finer microstructure at the annealing temperature (section 3.4). Thus, the smaller grain size of the TNB alloy seems to be responsible for its higher TIP-suszeptibility in comparison with the alloy TiAl35. This kind of microstructural effect on TIP was confirmed in a reference alloy γ-TAB, with the composition Ti-47Al-4(Cr, Mn, Nb, Si, B) which is characterized by a boron content of around 0.5at.%. Due to the pinning effect of the Titanium borides [25] this alloy is characterized by a fine colony size (50 μm) in the fully lamellar condition, indicating α-grains of this size at the annealing temperature of 1390 °C. The same alloy was produced without boron, designated as TABOB, and had a relatively coarse colony size of around 1 mm in the fully lamellar condition, similar to the binary alloy TiAl35. In terms of the grain size at 1390 °C the γ-TAB alloy resembles the present TNB alloy and the TABOB alloy resembles the present alloy TiAl35. Powders of these two reference alloys (γ-TAB and TABOB) were produced by the PIGA-technique and the particle size fractions 0–250 μm were filled into HIP-capsules, degassed, HIP′ ed at various temperatures (1000 and 1300 °C) and annealed at 1390 °C, 4 h for TIP-investigation. The Argon concentration of the HIP′ ed powder of TABOB was identical to that of the present PIGA-powder of the TNB alloy (Table 4, [17]). The results of the TIP-measurements after a high temperature exposure to 1390 °C are displayed in Fig. 7. Significant TIP was detected in the fine grained γ-TAB alloy, the porosity was determined to be 0.04vol.% (HIP 1000 °C) and 0.06vol.% (HIP 1300 °C), which is comparable to the porosity value of 0.05vol.% observed in TNB HIP′ ed at 1280 °C and annealed at 1390 °C for 4 h. In contrast, no distinct TIP was found in the boron-free counterpart TABOB. This result is regarded as further indication that a small grain size at the annealing temperature favors the occurence of TIP.

This correlation might be understood in the frame of the following senario: During atomization, inert gas is trapped in the liquid droplets. Since Ar and He are nearly unsoluble in metals, the inert gas is concentrated in gas-filled pores within the solidified powder particles. During subsequent HIP at 200 MPa and high temperature, the pores shrink while the pressure inside the pores increases. At the end of the compaction process, the temperature is reduced which results in a reduction of the inner pore pressure and an increase in the materials strength. In a second step, the pressure in the HIP chamber also is reduced. Due to the reduced inner pore pressure and the increased strength, the pores can not grow and the porosity remains below its detection level of 0.01%.

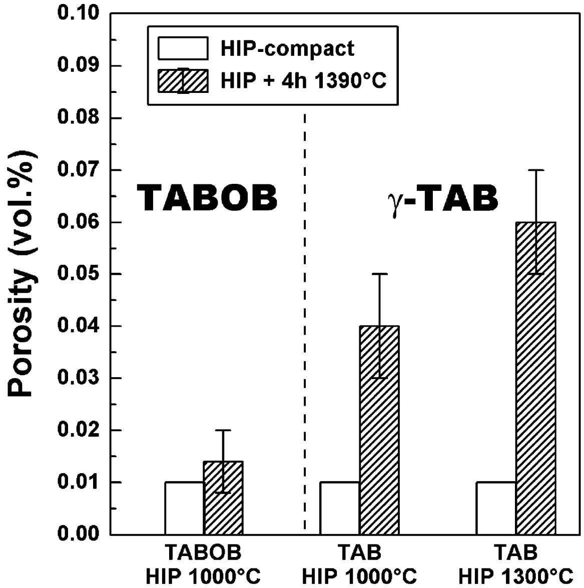

Fig. 7. Temperature induced porosity (TIP) evolution of two TiAl alloy powders (TABOB and TAB, powder size fraction 0–250 μm) consolidated by hot isostatic pressing (HIP) after a high temperature exposure to 1390 °C. The powders were produced by the PIGA-technique.

During a subsequent heat treatment at 1390 °C the inner pore pressure is again increased and the strength of the material is reduced. The pores thus grow with a related plastic deformation of the surrounding matrix. At the same high temperature a fine grained material has a lower strength compared with its coarser counterpart. Therefore, the resulting porosity is larger in fine grained material than in coarse grained material.

During the growth of a pore, the inner pore pressure decreases continuously which results in a decreasing rate of plastic deformation of the surrounding material. Therefore, an increase in annealing time causes an increase in the porosity level, however, an extension of the annealing time by a factor of 13 does not cause an increase in porosity by the same factor (Fig. 6).

4.2. Effect of the amount and type of entrapped inert gas in the powders

The inert gas concentration in the powders is another important factor for the occurrence of TIP in the heat treated powder compacts. In the case of the 'TIP-sluggish' (the term 'TIP-sluggish' is used here in terms of a coarse microstructure in the context of section 4.1) alloy TiAl35, significant TIP was only observed for Argon concentrations in excess of 2.0 μg/g (CA,Ar) and for long exposure times, but not for PIGA- and EIGA-processed TiAl35 with lower Argon concentrations (Table 4, Fig. 6a). For an intrinsically TIP-susceptible alloy with some grain-refining alloying additions, e.g. TNB, an Argon concentration of only 0.4 μg/g (PIGA) is already sufficient for TIP to occur after a long high temperature exposure to 1390 °C (4 h), but no TIP was seen for a short exposure time of 0.3 h. In the case of an Argon concentration in excess of 1.0 μg/g (TNB, EIGA) a high temperature exposure of only 0.3h already resulted in TIP evolution (Table 4, Fig. 6b). In this context the gas atomization technique should be recognised as the process which results in the least Argon entrapment, while the CA-technique gives the highest porosities (Table 2) and Argon concentrations (Fig. 4). This trend was previously seen in various studies on 304 stainless steel powders [19,20] and on Fe-40wt.%Ni-powder [26].

This study confirms that for gas atomized powders the porosity and the Argon concentration increase with increasing powder particle size in agreement with the results of previous studies [16–19,21]. Because the pores are filled with the processing gas, the inert gas concentration is closely connected to the porosity, which is confirmed in this study for gas atomized alloy powders (Table 2, Fig. 4). However, in the case of CA-powders no such correlation could be derived from the data. Numerous large powder particles produced by the CA-technique show cracks and a rupture-like appearance (Fig. 3c and d). For this reason the actual closed porosity may be significantly overestimated by the measuring method because an apparently closed pore may have an opening to the particle surface in a different section. Through such an opening the Argon gas would escape during the degassing procedure.

He atomized powders have a higher porosity compared with Ar atomized powders. Surprisingly, however, the porosity level after heat treatment of the compacted samples is much larger for the He than for the Ar containing material than would be expected from the initial powder porosity levels (Fig. 6). This seems not to be in accordance with the simple model (Section 4.1), since He and Ar should have a comparable influence on the TIP development as both are nearly ideal gases and insoluble in metals.

Therefore it might be speculated, that a diffusion controlled process could be responsible for the observed differences between He and Ar containing materials. The He atom is much smaller than the Ar atom and its diffusivity could be much higher. In addition, this assumption would not be in contradiction with the observed grain size effect, if the grain boundaries would serve as diffusion pathes for the gas.

On the other hand, for He atomized powders and related compacts, only porosity data and no He concentrations are available. Therefore it might be possible that the He gas concentration is much higher compared with that of Ar. If a large amount of the 'additional' He were stored in pores the dimensions of which are below the level of detection (1 μm) the submicron pores would also grow during subsequent heat treatment, become visible and could cause the observed large increase in porosity. This assumption is supported by the finding, that the porosity in the heat treated CA He-processed TNB compacts is much larger than the apparent porosity of the respective alloy powder (Fig. 6b). Thus, with the additional 'hidden He', the observed porosities could be rationalized within the simple model from section 4.1.

4.3. Aspects for application

With regard to a creep-resistant fully lamellar microstructure, the high temperature annealing time in the α single-phase or the a + β two-phase field for a real component will certainly be less than 1 h, probably in the range of 0.3 h dependent on the wall thickness. The long exposure time of 4 h within our experiments was simply employed to accentuate the effect of TIP evolution. No TIP was observed either for TiAl35 or for TNB after annealing for 0.3 h, when the powders were fabricated by gas atomization with the PIGA-technique using Argon gas. On the other hand mechanical stresses (tensile stresses are suspected to be most detrimental) exerted by hot working processes may exaggerate the tendency for TIP, and should be kept in mind.

From the results of this investigation four guidelines can be given for minimizing or completely preventing the occurrence of TIP: **1.** The inert gas concentration entrapped in the powder particles should be controlled carefully and kept at the lowest possible level. This is best accomplished by gas atomization with Argon, e.g. by the PIGA-technique, while centrifugal atomization results in higher inert gas concentrations in the powders. Moreover, the coarse powder size fractions in excess of e.g. 180 μm should be excluded in real application, since they contain a large amount of entrapped atomization gas. **2.** Argon as the atomizing medium is preferred over Helium. **3**. The annealing time should be kept as short as possible. **4**. An alloy with a minimum tolerable amount of grain refining addition, having a coarse grain size at the annealing temperature, is favorable to retard TIP.

5. Conclusions

1. Titanium aluminide and Titanium alloy powders produced by gas atomization of the melt by inert gas contain a certain amount of atomization gas, the concentration of which increases with the powder particle size.
2. Titanium aluminide alloy powders produced by centrifugal atomization in an inert gas atmosphere contain a higher amount of inert gas than the gas atomized powders. Among the gas atomization techniques investigated, the PIGA-technique gives the lowest inert gas concentrations over the whole powder particle size range investigated.
3. For minimizing the danger of TIP occurrence in real components fabricated from gamma Titanium aluminide alloys by a powder metallurgical route four guidelines can be given:
   - The amount of entrapped inert gas in the powder particles should be kept as low as possible, e.g. by gas atomization with the PIGA-technique and by avoiding the use of powder particles with sizes in excess of 180 μm for compaction.
   - For the atomizing medium Argon should be used rather than Helium.
   - The annealing time should be kept as short as possible.
   - An alloy should be selected, which has a large grain size at the annealing temperature.

Acknowledgements

The assistance of Dr. J.-X. Zhang from Shanghai Jiao Tong University with the porosity measurements is greatly acknowledged. The authors would also like to thank the technical staff H. Danker, F. Sasse, G. Müllauer for their contributions.

References

[1] Appel F, Brossmann U, Christoph U, Eggert S, Janschek P, Lorenz U et al. Advanced Engineering Materials. Wiley-VCH Weinheim 2000;2(11):699–720.
[2] Appel F, Wagner R. Mater. Sci. Engng 1998;R22:187–268.
[3] Clemens H, Kestler H. Advanced Engineering Materials. Wiley-VCH Weinheim 2000;2(9):551–70.
[4] Clemens H. Z. Metallkd 1995;86:814–22.
[5] Gerling R, Schimansky F-P, Wagner R. In: Aldinger F, editor. Materials by powder technology, PTM. Oberursel: DGM; 1993. p. 379–84.

[6] Gerling R, Schimansky FP, Wagner R. In: Capus JM, German RM, editors. Advances in powder metallurgy and particulate materials. Princeton, NJ: MPIF; 1992. p. 215–22.
[7] Liu CT, Maziasz PJ, Clemens DR, Schneibel JH, Sikka VK et al. In: Kim Y-W, Wagner R, Yamaguchi M, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1995. p. 679–86.
[8] Choi BW, Deng YG, McCullough C, Paden B, Mehrabian R. Acta. Metall. Mater 1990;38:2225–43.
[9] Semiatin SL. In: Kim Y-W, Wagner R, Yamaguchi M, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1995. p. 509–24.
[10] Habel U, Yolton CF, Moll JH. In: Kim Y-W, Dimiduk DM, Loretto MH, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1999. p. 301–8.
[11] Gerling R, Oehring M, Schimansky F-P, Wagner R. In: Phillips M, Porter J, editors. Advances in powder metallurgy and particulate materials, 3. Princeton: APMI; 1995. p. 91–9 Part 12.
[12] Kestler H, Clemens H, Baur H, Joos R, Gerling R, Cam G et al. In: Kim Y-W, Dimiduk DM, Loretto MH, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1999. p. 423–30.
[13] Gerling R, Bartels A, Clemens H, Oehring M, Schimansky F-P. Acta. Mater 1997;45:4057–66.
[14] Clemens H, Kestler H, Eberhardt N, Knabl W. In: Kim Y-W, Dimiduk DM, Loretto MH, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1999. p. 209–23.
[15] Clemens H, Glatz W, Schretter P, Yolton CF, Jones PE, Eylon D. In: Kim Y-W, Wagner R, Yamaguchi M, editors. Gamma titanium aluminides. Warrendale, PA: TMS; 1995. p. 55-5.
[16] Wegmann G, Gerling R, Schimansky FP, Zhang JX. In: Kosuge K, Nagai H, editors. Proc. of 2000 Powder Metallurgy World Congress. The Japan Society of Powder and Powder Metallurgy; 2000. p. 479–82.
[17] Gerling R, Leitgeb R, Schimansky FP. Mater. Sci. Engng 1998;A252:239–47.
[18] Gerling R, Schimansky F-P, Feddersen U. PM World Congress (1998), 46–51.
[19] Rabin BH, Smolik GR, Korth GE. Mater. Sci. Engng 1990;A124:1–7.
[20] Bae JC, Kelly TF, Flinn JE, Wright RN. Scr. Metall 1988;22:691–6.
[21] L'Estrade L, Hallén H, Ljunggren R. In: Gummeson PU, Gustafson DA, editors. Modern developments in powder metallurgy. Princeton, NJ: MPIF; 1988. p. 187–203.
[22] Eylon D, Schwenker SW, Froes FH. Metall. Trans 1985;16A:1526–31.
[23] Eylon D, Froes FH, Heggie DG, Blenkinsop PA, Gardiner RW. Met. Trans A 1983;14:2497–505.
[24] Eylon D, Froes FH. Titanium P/M Products. ASM Handbook. 1990 Formerly Tenth Edition, Metals Handbook, Vol. 2, pp. 647-660.
[25] Cheng TT. In: Kim DM, Dimiduk MH, Loretto S, editors. Gamma titanium aluminides. Warrendale, PA: TM; 1999. p. 389–96.
[26] Smolik GR, Delmore JE. Mater. Sci. Engng 1990;A124:15–9.